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In [6]:
```
import pandas as pd
```

In [7]:
```
x=pd.DataFrame([[1,2,3,4,5],[6,7,8,9,10]],index=[1,2],columns=["a","b","c","d","
x
```

Out[7]:

	a	b	c	d	e
1	1	2	3	4	5
2	6	7	8	9	10

In [8]:
```
x.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 2 entries, 1 to 2
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   a       2 non-null      int64
 1   b       2 non-null      int64
 2   c       2 non-null      int64
 3   d       2 non-null      int64
 4   e       2 non-null      int64
dtypes: int64(5)
memory usage: 96.0 bytes
```

In [9]:
```
x.describe()
```

Out[9]:

	a	b	c	d	e
count	2.000000	2.000000	2.000000	2.000000	2.000000
mean	3.500000	4.500000	5.500000	6.500000	7.500000
std	3.535534	3.535534	3.535534	3.535534	3.535534
min	1.000000	2.000000	3.000000	4.000000	5.000000
25%	2.250000	3.250000	4.250000	5.250000	6.250000
50%	3.500000	4.500000	5.500000	6.500000	7.500000
75%	4.750000	5.750000	6.750000	7.750000	8.750000
max	6.000000	7.000000	8.000000	9.000000	10.000000

In [10]:
```
x.isnull()
```

Out[10]:

	a	b	c	d	e
1	False	False	False	False	False
2	False	False	False	False	False

In [11]:
```
x.fillna(1)
```

Out[11]:

	a	b	c	d	e
1	1	2	3	4	5
2	6	7	8	9	10

In [12]:
```
x.duplicated()
```

Out[12]:
```
1    False
2    False
dtype: bool
```

In [13]:
```
x.drop_duplicates()
```

Out[13]:

	a	b	c	d	e
1	1	2	3	4	5
2	6	7	8	9	10

In [14]:
```
sum(x.duplicated())
```

Out[14]:
```
0
```

In [15]:
```
x.drop_duplicates()
```

Out[15]:

	a	b	c	d	e
1	1	2	3	4	5
2	6	7	8	9	10

In [16]:
```
y=pd.DataFrame({'Rollno':[101,102,103],'Name':['gaurav','aasd','asdfadsf']});
```

In [17]:
```
y
```

Out[17]:

	Rollno	Name
0	101	gaurav
1	102	aasd
2	103	asdfadsf

To Load the data from dataset

In [18]:
```
data=pd.read_csv('train_and_test2.csv')
```

In [19]:
```
data
```

Out[19]:

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	zero.4	...
0	1	22.0	7.2500	0	1	0	0	0	0	0	...
1	2	38.0	71.2833	1	1	0	0	0	0	0	...
2	3	26.0	7.9250	1	0	0	0	0	0	0	...
3	4	35.0	53.1000	1	1	0	0	0	0	0	...
4	5	35.0	8.0500	0	0	0	0	0	0	0	...
...	...	...	...	...	...	...	...	...	...	...	...
1304	1305	28.0	8.0500	0	0	0	0	0	0	0	...
1305	1306	39.0	108.9000	1	0	0	0	0	0	0	...
1306	1307	38.5	7.2500	0	0	0	0	0	0	0	...
1307	1308	28.0	8.0500	0	0	0	0	0	0	0	...
1308	1309	28.0	22.3583	0	1	0	0	0	0	0	...

1309 rows × 28 columns

Data Normalization

In [20]:

```
data.dtypes
```

Out[20]:

```
Passengerid      int64
Age            float64
Fare           float64
Sex              int64
sibsp            int64
zero             int64
zero.1           int64
zero.2           int64
zero.3           int64
zero.4           int64
zero.5           int64
zero.6           int64
Parch            int64
zero.7           int64
zero.8           int64
zero.9           int64
zero.10          int64
zero.11          int64
zero.12          int64
zero.13          int64
zero.14          int64
Pclass           int64
zero.15          int64
zero.16          int64
Embarked       float64
zero.17          int64
zero.18          int64
2urvived         int64
dtype: object
```

In [21]:
```
data['Passengerid']= data['Passengerid'].astype('object')
data['Sex']=data['Sex'].astype('object')
```

data prepared for label encoder

In [29]:
```
data['Sex'] = data['Sex'].apply(lambda x: 'Female' if x == 0 else 'Male')
```

In [30]:
```
data.head(5)
```

Out[30]:

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	zero.4	...	z
0	1	22.0	7.2500	Female	1	0	0	0	0	0	...	
1	2	38.0	71.2833	Male	1	0	0	0	0	0	...	
2	3	26.0	7.9250	Male	0	0	0	0	0	0	...	
3	4	35.0	53.1000	Male	1	0	0	0	0	0	...	
4	5	35.0	8.0500	Female	0	0	0	0	0	0	...	

5 rows × 28 columns

Use for label encoder

In [24]:
```
from sklearn import preprocessing
```

In [25]:
```
data['Sex'].unique()
```

Out[25]:
```
array(['Female', 'Male'], dtype=object)
```

In [26]:
```
label=preprocessing.LabelEncoder()
```

In [27]:
```
data['Sex']=label.fit_transform(data['Sex'])
```

In [28]:
```
data['Sex'].unique()
```

Out[28]:
```
array([0, 1])
```

Use of One-Hot Encoder

In [28]:
```
data['Sex'].unique()
```

Out[28]:
```
array(['Female', 'Male'], dtype=object)
```